

Toxin Production

Caused by *Staphylococcus aureus* and/or group A streptococcus toxin-producing bacteria.

Characteristic syndromes (all with skin manifestations) can be divided into three categories depending on the predominant clinical presentation:

- **Predominantly cutaneous** - Local: Bullous impetigo - Generalized: Staphylococcal scalded-skin syndrome
- **Intermediate cutaneous and systemic** - Local: Recurrent toxin-mediated erythema (recurrent perineal erythema)
- Generalized: - Recalcitrant erythematous desquamating disorder
 - Other abortive hybrids of predominantly systemic manifestations
- **Predominantly systemic**
- Generalized: - Toxic shock syndrome—*Staphylococcus* (predominantly) or *Streptococcus* - Scarlet fever—*Streptococcus* (predominantly) or *Staphylococcus*

PREDOMINANTLY CUTANEOUS

- Exfoliative toxins

Clinical Features

- Erythema (local or generalized)
- Tenderness
- Nikolsky sign
- Bullae
- Exfoliation

Foci of Bacterial Colonization/Infection

- Bullous impetigo - Local skin
- Staphylococcal scalded-skin syndrome - Conjunctiva - Middle ear - Nasopharyngeal - Umbilical stump - Carbuncle or boil

INTERMEDIATE CUTANEOUS AND SYSTEMIC

- Unidentified toxins

Clinical Features

- Erythema (generalized)
- Fever and hypotension
- Desquamation
- Mucosal involvement
- Prolonged disease
- Some but not all toxic shock syndrome criteria met
- Recurrences

Foci of Bacterial Colonization/Infection

- Recurrent perineal erythema
- Nasopharyngeal
- Recalcitrant erythematous desquamating disorder
- Non-HIV-associated - Surgical wounds
- Influenza
- Intravenous drug use
- Burn wounds
- Necrotizing fasciitis - Skin (in 80% of cases)
- Cellulitis
- Scarlet fever—streptococcal
- Surgical wounds

PREDOMINANTLY SYSTEMIC

- Superantigens

Clinical Features

- Erythema generalized with or without sandpaper characteristics
- Toxic appearance
- Fever and hypotension
- Mucosal involvement
- Swelling of hands and feet
- Organ failure
- Exfoliation or desquamation

Foci of Bacterial Colonization/Infection

- Toxic shock syndrome—*Staphylococcus aureus*
- Menstrual-associated
 - Vagina
- Non-menstrual
 - Sinusitis
 - Osteomyelitis
 - Gynecologic/postpartum
 - Surgical wounds
 - Skin (in 80% of cases)
 - Nasopharyngeal
 - Uterine
- Toxic shock syndrome—*Streptococcus*
- Necrotizing fasciitis